Solutions — Mock Olympiad, June 28, 2026

Problem 1. (IMC 2013, Day 2, Problem 1) Let $|z+1|>2$; prove $|z^3+1|>1$.

Solution. Since $z^3+1=(z+1)(z^2-z+1)$ and $|z+1|>2$, it suffices to prove $|z^2-z+1|\ge\frac12$. Write $z+1=re^{i\varphi}$ with $r=|z+1|>2$. Then

$$z^2-z+1=(re^{i\varphi}-1)^2-(re^{i\varphi}-1)+1=r^2e^{2i\varphi}-3re^{i\varphi}+3,$$

so

$$|z^2-z+1|^2=\big(r^2e^{2i\varphi}-3re^{i\varphi}+3\big)\big(r^2e^{-2i\varphi}-3re^{-i\varphi}+3\big).$$

Expanding and using $\cos 2\varphi=2\cos^2\varphi-1$,

$$|z^2-z+1|^2=r^4+9r^2+9-(6r^3+18r)\cos\varphi+6r^2(2\cos^2\varphi-1)=12\left(r\cos\varphi-\frac{r^2+3}{4}\right)^2+\frac14(r^2-3)^2.$$

The first term is ≥ 0, and since $r>2$ we have $(r^2-3)^2>1$, hence $|z^2-z+1|^2>\frac14$. Therefore $|z^2-z+1|>\frac12$ and $|z^3+1|=|z+1|\,|z^2-z+1|>2\cdot\frac12=1$. $\square$

Problem 2. (IMC 2012, Day 1, Problem 2) Smallest rank of an $n\times n$ matrix with zero diagonal and positive off-diagonal entries.

Solution. *Answer:* 0 for $n=1$, 2 for $n=2$, and $\boxed{3}$ for all $n\ge 3$.

For $n=1$ the matrix is (0), rank 0. For $n=2$ the matrix $\begin{pmatrix}0&b\\c&0\end{pmatrix}$ with $b,c>0$ has determinant $-bc<0$, so rank 2.

Lower bound for $n\ge3$: The first three rows are linearly independent. Suppose $c_1R_1+c_2R_2+c_3R_3=0$. Looking at column 1 (whose entry in row 1 is 0, the others positive), the relation $c_2a_{21}+c_3a_{31}=0$ with $a_{21},a_{31}>0$ forces c_2,c_3 to have opposite signs or both be zero. The same argument on columns 2 and 3 applies to the pairs (c_1,c_3) and (c_1,c_2). These three sign conditions can hold only if $c_1=c_2=c_3=0$. Hence rank ≥ 3.

Construction of rank 3: Take $M=\big((i-j)^2\big)_{i,j=1}^n$. The diagonal is 0 and off-diagonal entries are positive squares. Since

$$(i-j)^2=i^2\cdot1-2i\cdot j+1\cdot j^2,$$

M is the sum of three rank-one matrices (outer products $(i^2)(1)$, $(i)(j)$, $(1)(j^2)$), so $\operatorname{rank}M\le 3$. $\square$

Problem 3. (IMC 2013, Day 1, Problem 3) $2n$ students, n per trip; every pair must meet. Minimum number of trips.

Solution. *Answer:* 6 for every $n\ge2$.

Lower bound. In each trip a student meets $n-1$ others, so to meet all $2n-1$ classmates each student must go on at least 3 trips (since $2(n-1)<2n-1$). Thus the total attendance over all trips is at least $3\cdot 2n=6n$. Each trip contributes n attendances, so the number of trips is at least $6n/n=6$.

Construction with 6 *trips.* If n is even, split the $2n$ students into four equal groups A,B,C,D (size $n/2$ each) and run the trips

$$(A,B),\ (C,D),\ (A,C),\ (B,D),\ (A,D),\ (B,C).$$

Every pair of groups travels together once, so every two students meet.

If n is odd and divisible by 3, split into six equal groups E,F,G,H,I,J and run

$$(E,F,G),\ (E,F,H),\ (G,H,I),\ (G,H,J),\ (E,I,J),\ (F,I,J).$$

For the remaining n, write $n=2x+3y$ with $x,y\ge1$ (possible for every $n\ge2$ not covered above), form groups A,B,C,D of size x and E,F,G,H,I,J of size y, and superimpose the two schemes:

$$(A,B,E,F,G),\ (C,D,E,F,H),\ (A,C,G,H,I),\ (B,D,G,H,J),\ (A,D,E,I,J),\ (B,C,F,I,J).$$

In each case every pair of students shares a trip. $\square$

Problem 4. (IMC 2012, Day 1, Problem 4) If $f\in C^1(\mathbb{R})$ satisfies $f'(t)>f(f(t))$ for all t, prove $f(f(f(t)))\le0$ for $t\ge0$.

Solution. *Lemma 1. It is not the case that* $\lim_{t\to+\infty}f(t)=+\infty$. Suppose $f(t)\to+\infty$. Pick T_1 with $f(t)>2$ for $t>T_1$, then T_2 with $f(t)>T_1$ for $t>T_2$; for $t>T_2$ we get $f'(t)>f(f(t))>2$, so eventually $f(t)>t$. Then

$f'(t) > f(f(t)) > f(t)$, giving $f(t) > c\,e^{t}$ for large t. Feeding this back, $f'(t) > f(f(t)) > c\,e^{f(t)}$, so $f'(t)e^{-f(t)} > c$; integrating, the left side $e^{-f(T)} - e^{-f(t)}$ is bounded, while the right side grows without bound — contradiction.

Lemma 2. $f(t) < t$ for all $t > 0$. By Lemma 1 there exist arbitrarily large t with $f(t) < t$. If the claim failed, by continuity there is $t_0 > 0$ with $f(t_0) = t_0$. If some $t \geq t_0$ has $f(t) < t_0$, let $T = \inf\{t \geq t_0 : f(t) < t_0\}$; then $f(T) = t_0$ and $f'(T) > f(f(T)) = f(t_0) = t_0 > 0$, so $f > t_0$ just to the right of T, contradicting the definition of T. Otherwise $f(t) \geq t_0$ for all $t \geq t_0$, whence $f'(t) > f(f(t)) \geq t_0 > 0$ on (t_0, ∞), forcing $f(t) \to +\infty$ and contradicting Lemma 1.

Lemma 3. If $f(s_1) > 0$ and $f(s_2) \geq s_1$, then $f(s) > s_1$ for all $s > s_2$. In particular, if $s_1 \leq 0$ and $f(s_1) > 0$, then $f(s) > s_1$ for all $s > s_1$. If not, let $S = \inf\{s > s_2 : f(s) \leq s_1\}$, so $f(S) = s_1$ and $f'(S) > f(f(S)) = f(s_1) > 0$; this produces values $f < s_1$ to the left of S (if $S > s_2$) or $f > s_1$ to the right of s_2 (if $S = s_2$), each a contradiction. The second statement is the case $s_2 = s_1$.

Conclusion. Suppose, for contradiction, that some $t_0 > 0$ has $f(f(f(t_0))) > 0$. Put $t_1 = f(t_0)$, $t_2 = f(t_1)$, $t_3 = f(t_2) > 0$. Using Lemmas 2 and 3 one checks $t_1, t_2 > 0$: e.g. if $t_1 \leq 0$ then $f(t_1) \leq 0$ (otherwise Lemma 3(b) gives $f(t_0) > t_1$, impossible), so $t_2 \leq 0$, and then $t_3 = f(t_2) \leq 0$, contradicting $t_3 > 0$. Hence $0 < t_3 < t_2 < t_1 < t_0$ (the strict inequalities from Lemma 2). Now Lemma 3 gives $f(t) > t_1$ for $t > t_0$ and $f(t) > t_2$ for $t > t_1$, so for $t > t_0$,

$$f'(t) > f(f(t)) > t_2 > 0,$$

forcing $f(t) \to +\infty$ — contradicting Lemma 1. Therefore $f(f(f(t))) \leq 0$ for all $t > 0$, and the case $t = 0$ follows by continuity. □

Problem 5. (IMC 2012, Day 1, Problem 5) Prove $P(X) = X^{2^n}(X+a)^{2^n} + 1$ is irreducible over $\mathbb{Q}$, $a \in \mathbb{Q}$.

Solution. *Case $a = 0$.* Then $P(X) = X^{2^{n+1}} + 1$, whose roots are the primitive 2^{n+2}-th roots of unity; this is the cyclotomic polynomial $\Phi_{2^{n+2}}$, hence irreducible.

Case $a \neq 0$. Substitute $X = Y - \frac{a}{2}$. Then

$$P\big(Y - \tfrac{a}{2}\big) = \big(Y - \tfrac{a}{2}\big)^{2^n}\big(Y + \tfrac{a}{2}\big)^{2^n} + 1 = \Big(Y^2 - \tfrac{a^2}{4}\Big)^{2^n} + 1.$$

In the variable $Z = Y^2 - \frac{a^2}{4}$ this is $Z^{2^n} + 1 = \Phi_{2^{n+1}}(Z)$, again a cyclotomic (hence irreducible over $\mathbb{Q}$) polynomial; in particular it has no factor that is a polynomial in Y^2 other than constants and itself.

Suppose P is reducible; then so is $Q(Y) := \Big(Y^2 - \frac{a^2}{4}\Big)^{2^n} + 1$. Write $Q(Y) = \prod_i f_i(Y)^{m_i}$ with distinct monic irreducible f_i. Since $Q(Y) = Q(-Y)$ (it is a polynomial in Y^2), the multiset of factors is invariant under $Y \mapsto -Y$. No f_i is itself a polynomial in Y^2 (else it would give such a factor of $Z^{2^n} + 1$), so $f_i(-Y) \neq \pm f_i(Y)$ is impossible to fix each factor; the factors pair up as $f_{i'}(Y) = \pm f_i(-Y)$. Collecting one member of each pair gives a monic $f(Y)$ of degree 2^n with

$$Q(Y) = f(Y)\,f(-Y).$$

Let $b = f(0) \in \mathbb{Q}$. Comparing constant terms, $Q(0) = \Big(\frac{a^2}{4}\Big)^{2^n} + 1 = f(0)f(0) = b^2$. Setting $c = \big(\frac{a}{2}\big)^{2^{n-1}} \in \mathbb{Q} \setminus \{0\}$, this reads

$$c^4 + 1 = b^2.$$

But the Diophantine equation $c^4 + 1 = b^2$, i.e. $c^4 + 1^4 = b^2$, has no solution with $c \neq 0$: this is a special case of Fermat's theorem that $x^4 + y^4 = z^2$ has only trivial solutions (proved by infinite descent — a minimal positive solution yields, via the parametrization of primitive Pythagorean triples applied twice, a strictly smaller one). This contradiction shows P is irreducible. □